

Munesu Manyangarirwa

Data Scientist & Programme Operations Analyst

Based in Johannesburg, South Africa, I work at the intersection of data science, workforce analytics and programme operations, turning complex data into clear analysis, automated systems and decisions that move work forward.

With four years of continuous, hands-on experience, my work spans applied statistics, statistical modelling, SQL, Python, business intelligence, data visualisation and Google Cloud automation. My background in Industrial & Organisational Psychology, coupled with my current Honours studies at UNISA, brings a human-centred perspective to the way I analyse people, performance and organisational data.

About & Skills

My day-to-day work sits at the intersection of program operations and applied data science. I build automated analytics pipelines and business intelligence dashboards that program teams rely on for day-to-day decision-making, demonstrating my commitment to supporting team success, while maintaining and extending my statistical and machine-learning practice through dedicated data science projects.

This portfolio reflects both sides of that work: production analytics and applied data science, from operational reporting and automation to statistical analysis, machine learning and data visualisation.

PROGRAMMING & DATA

Python

SQL

Apps Script

pandas

NumPy

scikit-learn

Git

BI & VISUALISATION

Power BI

DAX

Looker Studio

Matplotlib

Seaborn

CLOUD & AUTOMATION

Google Cloud

Sheets API

Service Accounts

Pipeline Design

ANALYSIS & DOMAIN

Statistical Modelling

Classification

Workforce Analytics

Scope of work. A significant portion of my recent work, including ALX program operations data and confidential financial-sector analysis, is subject to NDA restrictions and cannot be published in full. Where applicable, I document the analytical approach, technical methods, and system architecture without exposing underlying datasets or proprietary information. This document, the linked website, and the GitHub repositories do not publish any real learner, client, or confidential financial data.

Four Years, Continuously

JUL 2024 →
PRESENT

Program Operations Associate

ALX Training SA · Remote

- Owned program operations across two programs and four concurrent cohorts, supporting 1,000+ learners through delivery, engagement, progression, and graduation
- Turned program data into operational intelligence, building automated reporting systems and dashboards that give teams a live view of learner performance, engagement, progression, and delivery
- Automated recurring operational workflows with Google Apps Script, reducing manual data processing by approximately 30 minutes per program per day and freeing program teams to focus on higher-value work
- Built and refined learner progression and graduation tracking systems, contributing to a 3 to 4 percentage-point increase in graduation rates across programs
- Connect data, people and program delivery, translating operational patterns into clear actions for program teams and stakeholders

JAN 2024 →
JUN 2024

Data Science & Data Analytics Technical Mentor

ALX Training SA · Remote

- Supported 1,000+ learners across Data Analytics and Data Science programs, combining technical guidance with data-driven learner support
- Engineered a Python-powered Looker Studio analytics pipeline, reducing daily data processing from 4 hours to 1.5 hours
- Delivered 10 live technical webinars, translating data science and analytics concepts into practical, learner-focused sessions
- Initiated a BigQuery migration proposal to improve the team's data infrastructure and reporting workflow

SEP 2023 →
DEC 2023

Junior Technical Mentor

ALX Training SA · Remote

- Provided data science technical support across the learner experience, resolving data and technical issues and coordinating webinar delivery with course leads
- Supported the delivery of technical data science learning sessions and contributed to program-level learner operations

Foundations & Education

JUL 2022 →
OCT 2022

Data Science Intern

ExploreAI · South Africa

- Applied statistical modelling and extreme-value analysis to HR and environmental datasets, developing analytical models for practical business and forecasting applications
- Built weather-forecasting models using Python and statistical methods
- Developed BI dashboards and executive-facing visualisations, translating analytical outputs into accessible business insights

MAR 2021 →
JUN 2021

Project Lead

North-West University

- Led a cross-disciplinary semester project from planning through delivery, coordinating team contributions and project execution
- Improved the final project score by 30% through structured planning, analysis, and delivery

EDUCATION

Honours, Industrial & Organisational Psychology

UNISA · in progress

BA, Industrial & Organisational Psychology

North-West University

Selected Projects

Learner Engagement Risk Model

Python · scikit-learn · 2026

A tool that predicts which learners are likely to disengage or drop out before it happens, so program teams can step in early. Built with machine learning trained on patterns like inactivity, quiz completion, and how quickly someone gets started, so support reaches the people who need it most.

- Full evaluation: accuracy, precision, recall, F1, ROC-AUC, and ROC curve comparison
- Class-weight balancing for the minority (churn) class; feature-importance analysis
- No real learner or client data used anywhere in this project

Program Analytics Automation Suite

Apps Script · Python · GCP

An automated system that replaces hours of manual spreadsheet work with dashboards that update themselves. It tracks learner pacing, payments, and progress across multiple training programs at once, saving the team real time every day.

- Cut manual reporting time by ~30 minutes per program per day; adopted beyond its original team
- Shown here as a generalised architecture description; production code is under NDA

Program Pacing & Engagement Analytics

Python · pandas · Seaborn

A recurring analysis that answers the questions program managers actually ask: who is falling behind, who is at risk of dropping off, and whether different learner groups are being served fairly. Turns raw activity data into clear, usable answers.

SQL & BI Work

Maji Ndogo Water Access Analysis

SQL · Power BI

A case study analysing water access across a country's provinces and towns, showing how raw records can be joined, cleaned, and ranked to reveal where infrastructure investment is needed most, then presented as an interactive map-based dashboard. Built and delivered as the technical mentor facilitating ExploreAI/ALX's relational-analysis curriculum.

HR People Analytics Dashboard

Power BI · DAX

A three-page dashboard that helps HR and people teams understand employee absenteeism and productivity: who is affected, why, and how workload relates to time away from work, so leaders can make better staffing and wellbeing decisions. Built with a direct line to my Industrial & Organisational Psychology background.

Power BI Teaching Demo: Company Financials

Power BI

A hands-on teaching dashboard designed from scratch to help learners practice real business intelligence skills, built around a company-financials scenario I created specifically for training purposes. All data is invented for instructional use.

2022 Foundations

Python · scikit-learn

Regression, classification, and unsupervised-learning sprint projects from my ExploreAI Data Science internship and certification, including extreme-value analysis applied to weather-event forecasting. The foundation the rest of this portfolio builds on.